

BINDU CHANDRA SHEKAR REDDY

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Professional Summary

Software Engineer and Computer Science graduate with experience in software development, testing, debugging, validation, and full-stack application development. Skilled in Java, Python, JavaScript, SQL, React, REST APIs, cloud technologies, and software engineering methodologies. Experienced in requirements analysis, technical documentation, production support, and cross-functional collaboration. Proven ability to deliver reliable software solutions throughout the Software Development Life Cycle (SDLC). Seeking to contribute technical and analytical expertise in software validation and engineering roles.

Skills

- **Programming Languages:** Java, Python, JavaScript, SQL, PHP, C
- **Software Engineering:** Software Development Life Cycle (SDLC), Testing & Validation, Debugging, Technical Documentation, Agile Development, Requirements Analysis, Code Review
- **Frameworks & Technologies:** React.js, Node.js, Spring Boot, REST APIs
- **Machine Learning:** Transformers, CNN, ANN, XGBoost, NumPy, Pandas, Matplotlib, TensorFlow/PyTorch, Scikit-learn
- **Cloud & DevOps:** AWS, Docker, Kubernetes, Google Kubernetes Engine (GKE)
- **Tools:** Git, VS Code, Google Colab, Xcode

Experience

California State University, Sacramento

Jan 2025 - May 2026

Instructional Student Assistant

Sacramento

- Assisted students in applying object-oriented programming concepts using Java and software engineering principles
- Guided students through debugging, testing, and validation of programming assignments
- Reviewed software projects and provided technical feedback on functionality and code quality
- Supported laboratory exercises using Git and VS Code

Teleindia Networks Pvt. Ltd

Jun 2023 - Jul 2024

Web Developer

Bengaluru, India

- Developed and maintained full-stack web applications using React, PHP, Node.js, and MySQL following SDLC best practices
- Performed application testing, debugging, and defect resolution to improve software reliability and user experience
- Collaborated with product managers, designers, and developers to gather requirements and deliver software solutions aligned with business objectives.
- Developed and integrated RESTful APIs and optimized backend services, reducing response times by 30%
- Supported deployment, monitoring, and troubleshooting of production systems, improving platform stability
- Created technical documentation for application features, APIs, and system enhancements

Nanorobotics Embed Technologies

Sep 2021 - Oct 2021

Machine Learning Intern

Bengaluru, India

- Developed and tested ANN models in Python with NumPy, Pandas and Matplotlib, achieving reliable predictions that supported downstream embedded analytics
- Programmed Python-based software modules for embedded devices and built ML inference pipelines, enabling real-time analytics for improved device performance.
- Presented findings on ML integration for embedded robotics to the engineering team, informing design decisions and advancing the project roadmap

Projects

NeuroFusionGPT – Multimodal Stress Detection System

Jul 2025 - May 2026

- Designed and implemented a multimodal software system for physiological stress detection using EEG and ECG datasets
- Developed an end-to-end machine learning pipeline including preprocessing, feature extraction, inference, and evaluation
- Conducted validation testing using accuracy, precision, recall, F1-score, and confusion matrix metrics
- Generated personalized AI-based wellness recommendations based on predicted stress levels

XML Transformation and Validation System

Feb 2026 - May 2026

- Contributed to the development and validation of XSLT transformations for XML-based data processing workflows
- Implemented and tested XSLT generation and validation components to ensure correct XML transformation outputs
- Performed debugging and validation testing to verify compliance with project requirements
- Collaborated with team members using software engineering best practices and version control tools
- Documented transformation logic, testing procedures, and validation results

IoT-Based Health Monitoring System with Edge Computing

Jan 2025 - May 2025

- Developed an iOS application for real-time health monitoring using wearable IoT sensor data
- Implemented edge computing functionality for low-latency data processing and anomaly detection

- Integrated AWS cloud services and containerized backend services using Docker and Kubernetes
- Conducted testing and validation of system functionality and data communication workflows

ML-based Medical Diagnostics for Infectious Diseases

Aug 2025 - Dec 2025

- Published at IEEE International Systems Conference (SysCon 2026), focusing on scalable ML-based diagnostic systems.
- Developed a supervised ML pipeline using epidemiological data (263K+ samples) for COVID-19 prediction.
- Engineered clinical and temporal features (comorbidity count, symptom-to-admission interval) to improve predictive performance.
- Built and optimized an XGBoost model achieving 0.80 ROC-AUC and 0.73 Recall for high-risk case detection.
- Designed a robust preprocessing pipeline to handle noisy, imbalanced real-world healthcare data.

Plant Leaf Disease Detection Using CNN

Aug 2021 - Jun 2022

- Developed a deep learning-based disease detection system using transfer learning techniques
- Achieved approximately 95% classification accuracy using VGG16 and ResNet50 architectures
- Built data preprocessing and visualization pipelines using Python and scientific computing libraries
- Enabled real-time disease classification for agricultural applications

Publications

- Bindu Chandra Shekar Reddy et al..ML-based Medical Diagnostics System for Infectious Diseases (Case Study: COVID-19).- *Published at IEEE International Systems Conference (SysCon 2026). Developed a machine learning-based diagnostic system using epidemiological data (263K+ samples). Built and optimized an XGBoost model achieving 0.80 ROC-AUC and 0.73 Recall for COVID-19 prediction. DOI: 10.1109/SysCon66367.2026.11503514*
- Bindu Chandra Shekar Reddy.Google Scholar Profile.*Full list of research publications available at: <https://scholar.google.com/citations?user=k67nfawAAAAJ>*

Education

California State University, Sacramento

Master of Science, Computer Science

Aug 2024 - May 2026

Sacramento, CA

KNS Institute of Technology (VTU)

Bachelor of Engineering, Computer Science and Engineering

Aug 2018 - Jun 2022

Bengaluru, India